

Deliverables for the Design Phase of the PGDip Capstone Project

(Due Date is Sunday of the first week of the New Term in September)

During the Design Phase of a project, producing clear and comprehensive evidence is crucial to ensure that the design decisions are well-documented, aligned with project requirements, and can be communicated effectively to stakeholders, particularly your supervisor. The evidence produced during this phase helps validate the feasibility, functionality, and viability of the project's design.

ALL the projects in PGDIP Course fall under the FIVE categories of the STREAMS: ***Application Development, Computer Networks, AI and Data Science, Web Technology, Information Systems***

Here are some key types of evidence that should be produced in the Design Phase of a project for each stream and submitted on blackboard:

Application Development Projects

1. Design Documents:

High-Level Design Document: An abstract portrayal of the system architecture, component interplay, and operational dynamics.

Detailed Design Document: A details of the design, dissecting modules, algorithms, and interfaces to unravel the complexities.

2. Diagrams and Visual Aids:

System Flowcharts: Pictorial depiction of data flows, interconnections, and operational pathways within the application.

Sequence Diagrams: Chronological blueprint of interactions among system components, actors, and their responses.

Entity-Relationship Diagrams: Graphical representation of data structures and relationships within the application.

3. Prototypes and Mockups:

Interactive Prototypes: Engaging models that simulate user interface interactions and grant stakeholders a tangible preview.

UI Mockups: Static representations, portraying the layout, aesthetics, and element arrangement within the application.

4. Requirements Traceability:

Links each design choice back to specific project requirements, ensuring a coherent synthesis of design and functionality.

RUBRIC

1. Design Documents: 30 marks

- This includes both High-Level and Detailed Design Documents. The abstract portrayal and detailed analysis of the system architecture, component interactions, and operational dynamics are crucial for laying the groundwork for development.

2. Diagrams and Visual Aids: 25 marks

- System Flowcharts, Sequence Diagrams, and Entity-Relationship Diagrams are pivotal for visualizing the system's structure and operations. These aids facilitate understanding and communication among team members and stakeholders.

3. Prototypes and Mockups: 25 marks

- Interactive Prototypes and UI Mockups play a key role in the design process, offering stakeholders a tangible preview of the application. They are crucial for validating design choices and user experience before full-scale development.

4. Requirements Traceability: 20 marks

- Ensuring that each design element is linked back to specific project requirements is essential for maintaining coherence and alignment with project objectives. This traceability is key for delivering a product that meets the defined needs.

Computer Networks Projects

1. Design Documents:

Network Architecture Blueprint: Structural schema detailing component hierarchy, interaction patterns, and connectivity.

Protocol Specification: Elucidation of communication protocols dictating data exchange and interaction among network entities.

2. Diagrams and Visual Aids:

Network Topology Diagrams: Visual representation outlining network nodes, their arrangement, and the pathways connecting them.

Data Flow Diagrams: Graphical depiction of data packet movement across the network, capturing routing and flow dynamics.

3. Prototypes and Mockups:

Network Simulations: Utilizing simulation tools to emulate network behavior, offering insights into potential performance metrics.

Interface Mockups: Displays of user interfaces for network management tools, shedding light on control mechanisms.

4. Requirements Traceability:

Establishes alignment between network design decisions and project requirements—ranging from performance to security.

RUBRIC

1. **Design Documents:** 30 marks

- This includes Network Architecture Blueprint and Protocol Specification. These documents are essential for detailing the structure, interaction patterns, and communication protocols of the network. They lay the foundational understanding for the entire network's operation.

2. **Diagrams and Visual Aids:** 25 marks

- Network Topology Diagrams and Data Flow Diagrams are crucial for visually mapping out the network. These aids are key for understanding the arrangement of network nodes, their connectivity, and the movement of data across the network.

3. **Prototypes and Mockups:** 25 marks

- Network Simulations and Interface Mockups are vital for pre-implementation testing and visualization. Simulations offer insights into potential network performance, while interface mockups provide a glimpse into the management tools' user interfaces.

4. **Requirements Traceability:** 20 marks

- Ensuring that network design choices align with project requirements is essential. This includes considerations for performance, security, scalability, and other crucial parameters. Traceability is key to fulfilling project objectives effectively.

AI and Data Science Projects

1. Design Documents:

Model Architecture Documentation: Thorough elucidation of AI model structure, algorithmic flow, and integration specifics.

Data Processing Workflow: Detailed exposition of data acquisition, cleaning, transformation, and model integration.

2. Diagrams and Visual Aids:

Model Flowcharts: Illustration of decision paths within AI models, demystifying the logic guiding predictive outcomes.

Data Flow Diagrams: Graphical representation of data movement, mapping transformations and interactions.

3. Prototypes and Mockups:

Model Prototypes: Dynamic models demonstrating predictive capabilities, granting stakeholders a tangible glimpse.

Data Visualization Mockups: Depictions of data insights through visual representations, offering clarity.

4. Requirements Traceability:

Bridges the gap between AI model objectives, data analysis goals, and design decisions shaping their embodiment.

RUBRIC

1. Design Documents: 30 marks

- Including Model Architecture Documentation and Data Processing Workflow. These documents are crucial as they provide a comprehensive overview of the AI model's structure, algorithmic flow, and data processing strategies. They lay the foundation for understanding the entire project's methodology and operation.

2. Diagrams and Visual Aids: 25 marks

- Model Flowcharts and Data Flow Diagrams are vital for visually explaining the AI models and the data journey. They aid in demystifying complex algorithms and data transformations, making the project more accessible and understandable.

3. Prototypes and Mockups: 25 marks

- Model Prototypes and Data Visualization Mockups are essential for demonstrating the practical application and effectiveness of the models. Prototypes give a real-world sense of the model's predictive capabilities, while visualization mockups help in presenting data insights clearly.

4. Requirements Traceability: 20 marks

- This aspect ensures that every design decision and model development aligns with the initial project objectives and data analysis goals. It's key for maintaining project coherence and ensuring that the final product meets the intended purpose.

Web Technology Projects

1. Design Documents:

Website Architecture Design: Layout showcasing page hierarchy, navigation patterns, and user interaction pathways.

API Specification: Documentation of endpoints, methods, and data structures facilitating web service interaction.

2. Diagrams and Visual Aids:

User Flow Diagrams: Visual roadmap illustrating user journey, interaction points, and navigation choices.

Wireframes: Schematic representation of web page layouts, highlighting the positioning of elements.

3. Prototypes and Mockups:

Interactive Prototypes: Engaging models simulating user interactions, providing stakeholders a firsthand experience.

UI Mockups: Static visual representations conveying design aesthetics, layout, and interface components.

4. Requirements Traceability:

Establishes a coherent connection between web design choices and the overarching project objectives and needs.

RUBRIC

1. Design Documents: 25 marks

- This includes Website Architecture Design and API Specification. The layout of page hierarchy, navigation patterns, user interaction pathways, and API details are fundamental for the project's structure and functionality. These documents provide a detailed blueprint for web development.

2. Diagrams and Visual Aids: 25 marks

- User Flow Diagrams and Wireframes are vital for visually planning the user's journey and the web page layouts. These tools are key in ensuring a user-centric design approach, mapping out interaction points and the visual arrangement of elements.

3. Prototypes and Mockups: 30 marks

- Interactive Prototypes and UI Mockups are essential for giving stakeholders a realistic preview of the web project. Interactive prototypes offer a tangible

experience of user interactions, while mockups convey the design aesthetics and layout, both critical for refining the final product.

4. Requirements Traceability: 20 marks

- Ensuring that all web design choices are coherently connected to the project's objectives and user needs is crucial. This aspect guarantees that the project remains aligned with its goals, fulfilling the intended purpose and requirements.

Information Systems Projects

1. Design Documents:

System Architecture Blueprint: Comprehensive layout detailing system components, their interconnections, and functions.

Database Schema: Structural depiction of the database design, encompassing tables, relationships, and attributes.

2. Diagrams and Visual Aids:

Data Flow Diagrams: Visual representation outlining data movement, transformation, and processing.

System Interaction Diagrams: Depicts interactions between system components and external entities.

3. Prototypes and Mockups:

System Prototypes: Functional models showcasing system behavior, allowing stakeholders to engage in user scenarios.

UI Mockups: Visual representations of user interfaces, aiding comprehension of design aesthetics and layout.

4. Requirements Traceability:

Establishes a clear linkage between design choices and overarching project requirements, catering to functionality and purpose.

RUBRIC

1. Design Documents: 30 marks

- Including System Architecture Blueprint and Database Schema. These documents are critical as they provide a detailed overview of the system's structure, component interrelations, and database design. They are foundational to the project's development and future scalability.

2. Diagrams and Visual Aids: 25 marks

- Data Flow Diagrams and System Interaction Diagrams are vital for visually conceptualizing the data movement and component interactions within the system. These aids are key in ensuring a clear understanding of the system's operational dynamics.

3. Prototypes and Mockups: 25 marks

- System Prototypes and UI Mockups play an important role in the development process. Prototypes allow stakeholders to interact with a functional model of the system, while UI mockups help in visualizing the interface design, both of which are critical for refining the final system.

4. Requirements Traceability: 20 marks

- This aspect ensures that every design and development decision aligns with the project's overarching requirements, including functionality, user needs, and purpose. It's key for maintaining project coherence and ensuring that the final product meets the intended objectives.